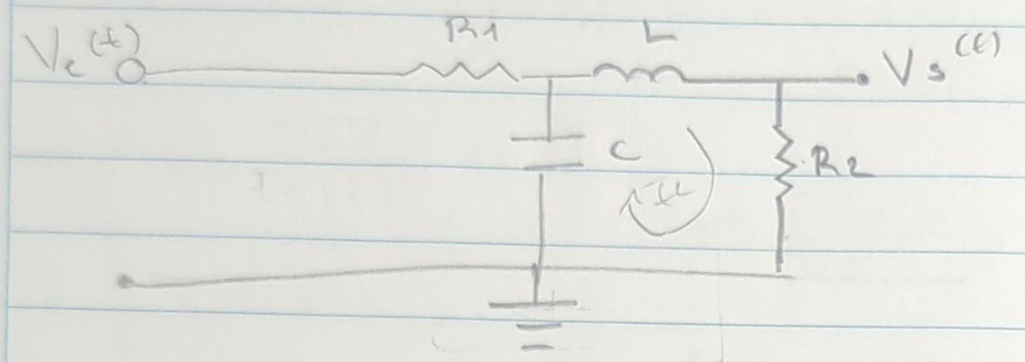
Análisis Matemático Sistema Endocrino   2 21-04-2026

Análisis matemático

1. Diagrama electrico



→ Modelo de ecuaciones Integro-diferenciales

1ra malla $V_e(t) = R_1 I_1(t) + \frac{1}{C}\int (I_1(t) - I_2(t))\,dt$

2da malla $0 = \frac{1}{C}\int (I_2(t) - I_1(t))\,dt + L\frac{dI_2(t)}{dt} + R_2 I_2 \longrightarrow \frac{1}{C}\int (I_1(t) - I_2(t))\,dt =$

$= L\frac{dI_2(t)}{dt} + R_2 I_2(t)$

→ Salida $V_s(t) = R_2 I_2(t)$

→ Transformada Laplace

$V_e(s) = R_1 I_1(s) + \frac{1}{Cs}(I_1(s) - I_2(s))$

$\frac{1}{Cs}(I_1(s) - I_2(s)) = (Ls + R_2) I_2(s)$

$I_1(s) - I_2(s) = Cs(Ls + R_2) I_2(s)$

$I_1(s) = (LCs^2 + CR_2 s + 1) I_2(s)$

→ Sustitución 1ra malla

$V_e(s) = R_1(LCs^2 + CR_2 s + 1) I_2(s) + (Ls + R_2) I_2(s)$

$V_e(s) = [(CLR_1)s^2 + (CR_1R_2 + L)s + (R_1 + R_2)] I_2(s)$

$V_s(s) = R_2 I_2(s) \longrightarrow I_2(s) = \frac{V_s(s)}{R_2}$

$V_e(s) = \frac{(CLR_1)s^2 + (CR_1 + R_2 + L)s + (R_1 + R_2)}{R_2} V_s(s)$

→ Función de transferencia

$$\frac{Vs(s)}{Ve(s)} = \frac{R2}{(CLR1)s^2 + (CR1R2 + L)s + (R1+R2)}$$

→ Error estado estacionario

$E(s) = Ve(s) - Vs(s) = Ve(s)\,[1 - FT]$

$E(s) = \frac{1}{s}(1 - FT)$

$$e_{(s)} = \lim_{s \to 0} s\,E(s)$$

$$e(s) = \lim_{s \to 0} \left(\frac{R2}{(CLR_1)s^2 + (CR_1R_2 + L)s + (R_1+R_2)} \right)$$

$$e(s) = \frac{R1}{R1+R2}$$

$$e\ s\ control = \frac{1k}{1k + 100k} = 0.0099$$

$$e\ s\ caso = \frac{1k}{1k+1k} = 0.5$$

→ Estabilidad del sistema de CONTROL

$$e_s = \frac{100 \times 10^3}{(1\times10^{-6} * 100\times10^{-3})s^2 + ((1\times10^{-6} * 1\times10^3 * 100\times10^3) + 1\times10^{-3})s + (1\times10^3 + 100\times10^3)}$$

$$= \frac{1}{1000}s^2 + \frac{100001}{1000}s + 101000$$

→ $s1 = -1011.012035$ → sistema estable con respuesta amortiguada

→ $s2 = -998998.988$

→ Estabilidad del sisteme CASO

$$es = \frac{1\times10^{3}}{(1000\times10^{-6} * 100\times10^{-3} \times 1\times10^{3})s^2 + ((1000\times10^{-6} * 1\times10^{3} * 1\times10^{3}) + 100\times10^{-3})s + (1\times10^{3} + 10^{3}}$$

$$= \frac{1}{10}s^2 + \frac{10004}{10}s + 1000000$$

→ $s_1 = -2.0002000\,6$ → sisteme estable respeto sobreamortiguado

→ $s_2 = -9999.9998$